Estimated Annual Agricultural Pesticide Use
Idaho, Statewide Ranking 2014-2018

21 July 2026

Summary

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides **statewide in Idaho** during 2018, as reported by USGS Estimated Annual Agricultural Pesticide Use data, summed across all Idaho counties. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification in `data/pesticide_classification.csv`, shared unchanged with the Washington pesticide class trends chapter). Rankings and estimates are based on the *EPest-high method* described by Thelin and Stone (2013) and Baker and Stone (2015):

> "EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD." *-Pesticide National Synthesis Project webpage*

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. "detailed interpretation of where and how much use occurs within a county is not appropriate"); summing county estimates to a statewide total inherits this caveat.
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 included once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
ALS	ALS-Inhibitor (Sulfonylurea & Related)
AVM	Avermectin
BIO	Biological/Microbial
BIP	Bipyridylium
CB	Carbamate
CAA	Chloroacetanilide
DMI	DMI Fungicide (Triazole/Imidazole)
DIA	Diamide
DNA	Dinitroaniline
DTC	Dithiocarbamate
FUM	Fumigant
GLY	Glycine/Phosphonate
INO	Inorganic/Mineral
IGR	Insect Growth Regulator

Table 1: Chemical class abbreviation key (used in the Class column of Table 2) *(continued)*

Code	Chemical class
NEO	Neonicotinoid
OC	Organochlorine
OP	Organophosphate
FUN	Other Fungicide
HRB	Other Herbicide
INS	Other Insecticide/Miticide
OTH	Other/Unclassified
PHX	Phenoxy
PHP	Phenylpyrazole
PGR	Plant Growth Regulator
PYR	Pyrethrin/Pyrethroid
SDH	SDHI
SPN	Spinosyn
QOI	Strobilurin (QoI)
TC	Thiocarbamate
TRZ	Triazine

Table 2: Top 80 EPest-high Estimates Statewide (ID), 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank (ID)	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	METAM POTASSIUM	FUM	6833506	3788663	2874904
2	METAM	FUM	4809535	5455738	5993341
3	CHLOROPICRIN	FUM	2031809	870762	529045
4	SULFURIC ACID	INO	1598628	3313187	3558072
5	DICHLOROPROPENE	FUM	1367078	1493001	1508878
6	GLYPHOSATE	GLY	888738	765951	719412
7	BACILLUS AMYLOLIQUEFACIEN	BIO	418964	199800	199800
8	2,4-D	PHX	214705	153963	131233
9	EPTC	TC	174666	220510	230709
10	PENDIMETHALIN	DNA	168211	174652	148241
11	CHLOROTHALONIL	FUN	138764	139496	131277
12	METRIBUZIN	TRZ	124641	121968	119441
13	MCPA	PHX	124519	129273	149175
14	METOLACHLOR & METOLACHLOR-S	CAA	100012	94964	105685
15	DIMETHENAMID & DIMETHENAMID-P	CAA	93802	119626	100467
16	DIMETHENAMID-P	CAA	93802	91402	83532
17	MANCOZEB	DTC	89213	92481	138211
18	METOLACHLOR-S	CAA	81295	74582	78570
19	BROMOXYNIL	HRB	77415	65040	87066
20	ACETOCHLOR	CAA	74124	34616	40102
21	FLUROXYPYR	HRB	56387	58028	55439
22	CHLORPYRIFOS	OP	56147	67094	66627
23	MALEIC HYDRAZIDE	PGR	54238	36646	46260
24	SULFUR	INO	52846	78970	88766
25	ATRAZINE	TRZ	47753	107971	69063
26	METHOMYL	CB	44764	38918	32844
27	PHOSPHORIC ACID	INO	42355	49265	82250
28	PARAQUAT	BIP	41906	44064	52485
29	TRIFLURALIN	DNA	31919	29904	29050
30	PYRIMETHANIL	FUN	29914	23966	22363
31	PETROLEUM OIL	INS	29102	314695	201738
32	BURKHOLDERIA SPP	BIO	28801	42696	42696
33	DICAMBA	HRB	27062	27833	23448
34	ETHALFLURALIN	DNA	25271	51365	45270
35	DIQUAT	BIP	24545	22251	23314

Table 2: Top 80 EPest-high Estimates Statewide (ID), 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) *(continued)*

Rank (ID)	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
36	AZOXYSTROBIN	QOI	23650	20131	19412
37	BOSCALID	SDH	23035	27745	26335
38	HYDROGEN PEROXIDE	INO	23019	11510	7681
39	HEXAZINONE	TRZ	22593	23937	19807
40	QUINTOZENE	FUN	21783	14562	14007
41	PYRACLOSTROBIN	QOI	20800	22859	24312
42	PROPARGITE	INS	19473	19473	37578
43	THIOPHANATE-METHYL	FUN	19143	20668	15090
44	PROPICONAZOLE	DMI	19059	19464	18444
45	METOLACHLOR	CAA	18717	20383	27116
46	GLUFOSINATE	GLY	17424	9461	7360
47	DIURON	HRB	16884	32861	27993
48	IMIDACLOPRID	NEO	15468	17245	20780
49	TRI-ALLATE	TC	15088	15081	17933
50	OXAMYL	CB	14787	9487	34929
51	PINOXADEN	HRB	14712	14072	14059
52	CLETHODIM	HRB	12884	10244	9160
53	FLUOPYRAM	SDH	12635	9874	8328
54	DCPA	HRB	12022	20827	15818
55	PICLORAM	HRB	11668	4172	2818
56	DIMETHOATE	OP	11564	36457	41898
57	CLOPYRALID	HRB	11049	13694	13587
58	HEXYTHIAZOX	INS	10040	10128	9109
59	BENTAZONE	HRB	9412	6476	8670
60	FLUXAPYROXAD	SDH	8227	8215	6569
61	CALCIUM POLYSULFIDE	INO	8134	4560	3639
62	MEFENOXAM	FUN	6993	6887	6558
63	PICOXYSTROBIN	QOI	5981	3057	2902
64	BIFENTHRIN	PYR	5649	4266	5015
65	PROTHIOCONAZOLE	DMI	5623	2631	1760
66	TEBUCONAZOLE	DMI	5446	6264	4453
67	DIFENOCONAZOLE	DMI	4847	4345	5988
68	SAFLUFENACIL	HRB	4226	3811	4145
69	FLUTOLANIL	SDH	4172	14248	14531
70	SPIROTETRAMAT	INS	3855	3258	2514
71	CYHALOTHRIN-LAMBDA	PYR	3721	4569	4146
72	CARFENTRAZONE-ETHYL	HRB	3669	2491	1803
73	FLUAZINAM	FUN	3602	3904	6008
74	MCPB	PHX	3600	1821	1943
75	MANDIPROPAMID	FUN	3448	2696	3099
76	METIRAM	DTC	3411	3411	7878
77	CYFLUTHRIN	PYR	3361	1997	1445
78	THIFENSULFURON	ALS	3334	3263	2771
79	ALPHA CYPERMETHRIN	PYR	3324	2690	1799
80	MALATHION	OP	3162	7436	8486

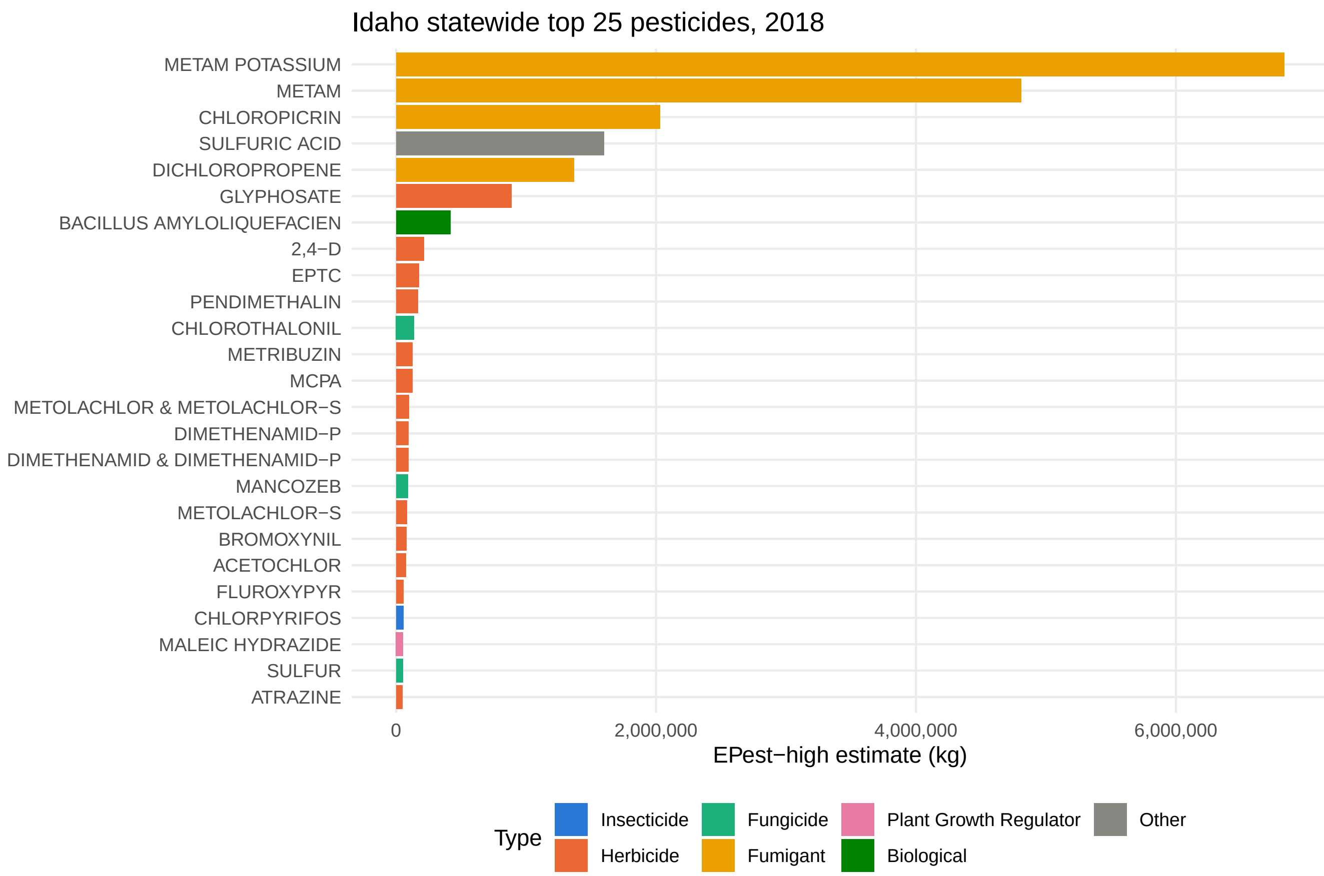

Figure 1: Top 25 compounds statewide by EPest-high mass applied, most recent year, colored by pesticide type.

Appendices

Table 3: Top 50 Statewide Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM POTASSIUM	6833506	6833506
2	METAM	4809535	4809535
3	CHLOROPICRIN	508044	2031809
4	SULFURIC ACID	1463266	1598628
5	DICHLOROPROPENE	1367078	1367078
6	GLYPHOSATE	795760	888738
7	BACILLUS AMYLOLIQUEFACIEN	1874	418964
8	2,4-D	189776	214705
9	EPTC	122637	174666
10	PENDIMETHALIN	118240	168211
11	CHLOROTHALONIL	138764	138764
12	METRIBUZIN	119227	124641
13	MCPA	124519	124519
14	METOLACHLOR & METOLACHLOR-S	58359	100012
15	DIMETHENAMID & DIMETHENAMID-P	72676	93802
16	DIMETHENAMID-P	72676	93802
17	MANCOZEB	72481	89213
18	METOLACHLOR-S	51999	81295
19	BROMOXYNIL	76989	77415
20	ACETOCHLOR	37608	74124
21	FLUROXYPYR	55859	56387
22	CHLORPYRIFOS	53713	56147
23	MALEIC HYDRAZIDE	15082	54238
24	SULFUR	33561	52846
25	ATRAZINE	1101	47753
26	METHOMYL	9930	44764
27	PHOSPHORIC ACID	30568	42355
28	PARAQUAT	9173	41906
29	TRIFLURALIN	2003	31919
30	PYRIMETHANIL	29914	29914
31	PETROLEUM OIL	29102	29102
32	BURKHOLDERIA SPP	120	28801
33	DICAMBA	11076	27062
34	ETHALFLURALIN	25271	25271
35	DIQUAT	22866	24545
36	AZOXYSTROBIN	22038	23650
37	BOSCALID	16920	23035
38	HYDROGEN PEROXIDE	960	23019
39	HEXAZINONE	4603	22593
40	QUINTOZENE	5315	21783
41	PYRACLOSTROBIN	20393	20800
42	PROPARGITE	991	19473
43	THIOPHANATE-METHYL	11926	19143
44	PROPICONAZOLE	18156	19059
45	METOLACHLOR	6360	18717
46	GLUFOSINATE	16082	17424
47	DIURON	4807	16884
48	IMIDACLOPRID	13532	15468
49	TRI-ALLATE	12106	15088
50	OXAMYL	9497	14787

Table 4: Top 50 Statewide Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	5356705	5455738
2	METAM POTASSIUM	3663951	3788663
3	SULFURIC ACID	3127033	3313187
4	DICHLOROPROPENE	1485004	1493001
5	CHLOROPICRIN	267757	870762
6	GLYPHOSATE	725324	765951
7	PETROLEUM OIL	36088	314695
8	ETHOPROPHOS	0	251521
9	EPTC	160265	220510
10	PROPAMOCARB HCL	0	219497
11	BACILLUS AMYLOLIQUEFACIEN	4613	199800
12	PENDIMETHALIN	127265	174652
13	2,4-D	136117	153963
14	CHLOROTHALONIL	138952	139496
15	MCPA	126488	129273
16	METRIBUZIN	117256	121968
17	DIMETHENAMID & DIMETHENAMID-P	86410	119626
18	ATRAZINE	31342	107971
19	METOLACHLOR & METOLACHLOR-S	54921	94964
20	MANCOZEB	83680	92481
21	DIMETHENAMID-P	74402	91402
22	DIMETHENAMID	36022	84672
23	SULFUR	27464	78970
24	METOLACHLOR-S	47187	74582
25	CHLORPYRIFOS	62995	67094
26	BROMOXYNIL	63253	65040
27	FLUROXYPYR	53363	58028
28	ETHALFLURALIN	28587	51365
29	PHOSPHORIC ACID	44205	49265
30	PARAQUAT	7988	44064
31	BURKHOLDERIA SPP	4392	42696
32	METHOMYL	7016	38918
33	MALEIC HYDRAZIDE	9434	36646
34	DIMETHOATE	18823	36457
35	ACETOCHLOR	18203	34616
36	DIURON	7615	32861
37	TRIFLURALIN	2949	29904
38	DICAMBA	18033	27833
39	BOSCALID	21778	27745
40	NALED	3605	26552
41	PYRIMETHANIL	23966	23966
42	HEXAZINONE	8765	23937
43	PYRACLOSTROBIN	21738	22859
44	DIQUAT	20387	22251
45	DCPA	0	20827
46	THIOPHANATE-METHYL	10422	20668
47	METOLACHLOR	7734	20383
48	AZOXYSTROBIN	18197	20131
49	PROPARGITE	991	19473
50	PROPICONAZOLE	18058	19464

Table 5: Top 50 Statewide Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	5789258	5993341
2	SULFURIC ACID	3342612	3558072
3	METAM POTASSIUM	2700520	2874904
4	DICHLOROPROPENE	1349724	1508878
5	GLYPHOSATE	689643	719412
6	CHLOROPICRIN	167015	529045
7	EPTC	166519	230709
8	PROPAMOCARB HCL	0	219497
9	PETROLEUM OIL	34142	201738
10	BACILLUS AMYLOLIQUEFACIEN	4613	199800
11	ETHOPROPHOS	0	171342
12	MCPA	146692	149175
13	PENDIMETHALIN	115358	148241
14	MANCOZEB	132941	138211
15	CHLOROTHALONIL	130851	131277
16	2,4-D	116520	131233
17	METRIBUZIN	114451	119441
18	METOLACHLOR & METOLACHLOR-S	54662	105685
19	DIMETHENAMID & DIMETHENAMID-P	74820	100467
20	SULFUR	34798	88766
21	BROMOXYNIL	73064	87066
22	DIMETHENAMID	36022	84672
23	DIMETHENAMID-P	67616	83532
24	PHOSPHORIC ACID	62649	82250
25	METOLACHLOR-S	44282	78570
26	ATRAZINE	21027	69063
27	CHLORPYRIFOS	61939	66627
28	FLUROXYPYR	51099	55439
29	PARAQUAT	8939	52485
30	MALEIC HYDRAZIDE	13269	46260
31	ETHALFLURALIN	31338	45270
32	BURKHOLDERIA SPP	4392	42696
33	DIMETHOATE	19500	41898
34	ACETOCHLOR	21730	40102
35	PROPARGITE	4860	37578
36	OXAMYL	30929	34929
37	METHOMYL	7578	32844
38	TRIFLURALIN	3018	29050
39	DIURON	7986	27993
40	METOLACHLOR	10380	27116
41	NALED	3605	26552
42	BOSCALID	22377	26335
43	PYRACLOSTROBIN	21679	24312
44	DICAMBA	15345	23448
45	DIQUAT	22195	23314
46	PYRIMETHANIL	22363	22363
47	IMIDACLOPRID	20343	20780
48	HEXAZINONE	8201	19807
49	AZOXYSTROBIN	17418	19412
50	PROPICONAZOLE	16617	18444